

1.0 Scope	2.0 Resource Required
1.1 Purpose: To ensure effective management of rolling machine rollers through digital tracking, storage, inspection control, vendor management, approval workflow, dashboard monitoring, inventory monitoring, usage & disposal. 1.2 Area of Application: Applicable to all rolling machine rollers 1.3 Responsibility: Production, Purchase, Engineering & Quality engineer. Roller management incharge is responsible to update roller status in RMS which is being approved by HOD – Hot end. 1.4 Accountability: Production Head	2.1 Man: Engineering team, Quality person, Purchase executive, Production Person. 2.2 Machine: Roughness tester, dial gauge, measuring tape, Crane / Forklift (for handling), Lathe machine, micrometer, Chrome thickness tester & texture measurement instrument 2.3 Material: Rollers (Main Roller), Wooden saddle supports, Protective covers.
3.0 Terms and Definition	4.0 Key Performance Requirements
3.1 RMS: Roller Management System used for digital tracking and records 3.2 PDI: Pre-Dispatch Inspection 3.3 NCR: Non-Conformance Report	4.1 Quality: Roller roughness shall be within specified limits, Roller Ra, and visual quality to meet defined standards, shaft Ra, run-out, chrome thickness, roller diameter, shaft diameter. 4.2 EHS: Safe lifting and handling of rollers, Use of certified lifting tools only, PPE compliance mandatory
5.0 Frequency	6.0 Competency Enhancement
5.1 Inspection of used rollers: All rollers 5.2 Inspection of engraved roller at vendor: All rollers (using ROM/L4/006) 5.2 PDI of roller (post-chrome) at vendor: All rollers. Inspection report (in RMS & ROM/L4/005) 5.3 Incoming inspection at BRL: All rollers 5.4 Inspection during mounting: All rollers.	6.1 Inspection training 6.2 Job card preparation training 6.3 RMS updating training 6.4 SOP training

	Prepared By	Reviewed By	Approved By	Page number
Signature				Page 1 of 12
Name	Mr. Ripal Rana	Mr. Prabhakar P	Mr. Ajay Singh	
Designation	Sr. Executive	HOD - Hotend	Plant head	

7.0 Procedure

7.1 Sequence of Operation

7.1.1 Roller Storage Location:

a. The roller shall be stored to the designated roller storage area; the storage location is as below:

1. Incoming roller from vendor.
2. Inspected rollers were found OK.
3. Used rollers.
4. Ready to use – near rolling machine (1-set of rollers).

b. Storage shall be categorized based on:

- Roller Diameter (260, 340 & 400 mm)
- Roller type (top or bottom)
- Status (Ready/under inspection/used).

c. In Storage location Room shall be color-coded and clearly marked for:

- Ready for use - Green
- Incoming roller from vendor – Yellow
- Used rollers - Orange
- Rejected / to be scraped - Red

d. The production team shall ensure that each roller is:

- Properly identified with Roller Number.
- Tagged with status in RMS (Ready / Installed / Vendor / Scrap)
- History card updation to be done for each movement for all rollers.

Note: Rollers shall be stored in horizontal position only, with adequate spacing to avoid accidental contact or impact.

e. Direct contact of roller surface with floor, metal, or other rollers is strictly prohibited.

f. For long-term storage - protective wrapping shall be applied.

	Prepared By	Reviewed By	Approved By	Page number
Signature				Page 2 of 12
Name	Mr. Ripal Rana	Mr. Prabhakar P	Mr. Ajay Singh	
Designation	Sr. Executive	HOD - Hotend	Plant head	

- g. Only authorized personnel shall be allowed to handle or shift rollers from storage.

Note: Rollers shall not be issued for installation unless storage condition is satisfactory and RMS status is updated as "Ready to Use".

7.1.2 Roller Removal & Repair Decision:

- a. Roller removal shall be initiated based on visual quality issues, roughness deviation or planned replacement. Roller shall be removed as per rolling machine SOP.
- b. Machines shall be stopped and isolated as per safety procedures before removal.
- c. Roller removal should be done using approved lifting tools only and kept at **Used rollers** storage location
- d. After removal, the roller shall be cleaned with sand blasting (1 cut or more if required) and visually jointly inspected by Engineering team and Production team.
- e. Detailed inspection shall be carried out covering:
 - Surface condition and chrome integrity
 - Roughness value
 - Engraving depth
 - Shaft runout
 - Bearing seat condition
- f. All inspection results shall be recorded in the [RMS](#) Web app along with remarks.
- g. Based on inspection results, repair decisions shall be taken jointly by Engineering & Production team (Roller incharge), a job card will be prepared through [RMS](#) as well as using Job card format (ROM/L4/004).
 - Re-chrome – if surface damage or chrome wear is within repairable limit
 - Re-engraving – if engraving depth (roughness) or pattern is below specification
 - Shaft repair – if shaft or journal damage is repairable
 - Scrap – if roller exceeds scrap cycle limit or damage is beyond repair

	Prepared By	Reviewed By	Approved By	Page number
Signature				Page 3 of 12
Name	Mr. Ripal Rana	Mr. Prabhakar P	Mr. Ajay Singh	
Designation	Sr. Executive	HOD - Hotend	Plant head	

h. Scrap decision shall be approved by the Production Head, Plant Head, CFO and CEO. It should be recorded in RMS. Scrap decision shall be taken if roller is:

- Roller is Cracked.
- Diameter is less than mentioned in specification
- Roller is bend.

i. RMS status shall be updated accordingly:

- Under Repair
- At Vendor
- Scrap

Note: No roller shall be sent to vendor or scrapped without inspection record and RMS approval.

7.1.3 Vendor Authorization & Repair Time:

- a. Only authorized vendors shall be permitted to carry out repair activities on Rollers.
- b. Vendor authorization shall be granted based on:
 - Technical capability to repair rollers
 - Availability of required machinery, tools, and calibrated instruments
 - Skilled and trained manpower
 - Past performance in terms of quality, turnaround time, and compliance
- c. Each roller sent for repair shall be accompanied by an approved job card (in [RMS](#) & ROM/L4/004)
- a. Vendor repair performance shall be periodically reviewed based on:
 - Job card compliance
 - Repair turnaround time
 - PDI results
 - Repeat failures
- b. Vendors failing to meet job card requirements or repair timelines shall be:
 - Reviewed
 - Put under observation

	Prepared By	Reviewed By	Approved By	Page number
Signature				Page 4 of 12
Name	Mr. Ripal Rana	Mr. Prabhakar P	Mr. Ajay Singh	
Designation	Sr. Executive	HOD - Hotend	Plant head	

- Removed from authorized vendor list if required
- c. All vendor authorization records, job cards, and performance evaluations shall be maintained.

7.1.4 Transportation BRL Plant to Vendor:

- a. Transportation of rollers from BRL plant to vendor shall be done only after completion of inspection and approval for repair or rework and defined in Job card (ROM/L4/004).
- b. Before dispatch, the roller shall be properly cleaned with sand blasting, wrapped with plastic film and packed in wooden box to avoid damage during transit.
- c. Packing shall be included.
 - Wooden saddle supports
 - Rubber padding at all contact points
 - Waterproof and dust-proof covering
- d. Roller surfaces shall not meet vehicle floors, metal structures, or other rollers.
- e. Suitable vehicles shall be selected based on roller size and weight to ensure safe transportation without vibration or impact.
- f. Rollers shall be properly restrained inside the vehicle to prevent rolling, shifting, or collision during transit.
- g. Dispatch shall be done only to approved vendors as per purchase and quality requirements.
- h. Dispatch documents shall include:
 - Delivery challan
 - Job card
 - Defined repair scope and special instructions (if any)
- i. RMS status shall be updated as "Sent to Vendor."
- j. Loading and unloading shall be carried out using crane or forklift with nylon slings and wooden saddles only.

Note: Manual handling, rolling on floor, or dragging of rollers is strictly prohibited.

	Prepared By	Reviewed By	Approved By	Page number
Signature				Page 5 of 12
Name	Mr. Ripal Rana	Mr. Prabhakar P	Mr. Ajay Singh	
Designation	Sr. Executive	HOD - Hotend	Plant head	

- k. Any damage observed during loading or transit shall be recorded and communicated to vendor and Purchase, Production and Quality team immediately.

7.1.5 Post-Laser Engraving inspection of Roller:

- a. Once the vendor is done with the engraving and other repairing activity, they will communicate to BRL team for the Post-Laser Engraving inspection before starting the Chrome coating on the roller surface

- b. Based on communication, BRL team will plan the Post-Laser Engraving inspection and BRL representative conduct the Post-Laser Engraving inspection.

Note: Inspection team must include the person from production, quality team.

- c. Post-Laser Engraving inspection will be conducted based on the parameter & tolerance mentioned in the Inspection plan (ROM/L4/009).

Note: All the instruments and equipment which will be used in Inspection must be calibrated.

- d. All the observed & measured values need to be noted in the Post-Laser Engraving inspection checklist (ROM/L4/006).

- e. In case of satisfactory results (mentioned in Spec. and other visual & mechanical condition) complete the Inspection. And signed it as **"APPROVED."**

- f. In case of deviation:

- If deviation is tolerable, approve by Plant head with deviation note ROM/L4/008 (refer Annexure 1).
- If deviation is critical and can effect on glass quality, stamp as **"REJECTED."** And NCR (ROM/L4/007) will be initiated by BRL.

- g. All NCs will be recorded and NCR to be filled & submitted by supplier for listed NCs.

- h. Based on the NC and corrective action, supplier will do the correction

- i. Follow step c., d. & e once the rework is done by supplier

7.1.6 PDI of Roller:

- a. Once the vendor is ready with the Chrome coated roller, they will communicate with the BRL team through mail as they are ready to have PDI.

- b. Based on communication, BRL team will plan the PDI at the vendor place.

Note: PDI team must include the person from production, quality, Engineering team.

	Prepared By	Reviewed By	Approved By	Page number
Signature				Page 6 of 12
Name	Mr. Ripal Rana	Mr. Prabhakar P	Mr. Ajay Singh	
Designation	Sr. Executive	HOD - Hotend	Plant head	

- c. PDI will be conducted based on the parameter & tolerance mentioned in the Inspection plan (ROM/L4/009).

Note: All the instruments and equipment which will be used in PDI must be calibrated.

- d. All the observed & measured values need to be noted in the Roller PDI checklist (ROM/L4/005) along with the deviation and remarks (in RMS).
- e. In case of satisfactory results (mentioned in Spec. and other visual & mechanical condition) complete the PDI. And signed it as **“APPROVED.”**
- f. In case of deviation:
- If deviation is tolerable, approve by Plant head with deviation note ROM/L4/008 (refer Annexure 1).
 - If deviation is critical and can effect on glass quality, stamp as **“REJECTED.”** And NCR (ROM/L4/007) will be initiated by BRL.
- g. All NCs will be recorded and NCR to be filled & submitted by supplier for listed NCs.
- h. Once the rework is done, follow step c., d. & e.

Note: Dispatch should be done only after PDI Approved.

7.1.7 Transportation Vendor to BRL Plant:

- a. Transportation of rollers from vendor to BRL shall be carried out only after successful completion of PDI.
- b. Vendor should ensure that rollers are packed properly before dispatching:
- Wooden saddle supports
 - Rubber padding on contact points
 - Waterproof and dust-proof covering
- c. Suitable vehicles should be selected based on roller size and weight to prevent movement, vibration, or impact during transit.
- d. Rollers shall be adequately restrained inside the vehicle to avoid rolling or shifting.
- e. Dispatch documents shall include:
- Approved PDI checklist (ROM/L4/005 & ROM/L4/006)

	Prepared By	Reviewed By	Approved By	Page number
Signature				Page 7 of 12
Name	Mr. Ripal Rana	Mr. Prabhakar P	Mr. Ajay Singh	
Designation	Sr. Executive	HOD - Hotend	Plant head	

- Delivery challan
 - NCR with corrective action (if applicable)
- f. Any deviation, damage, or abnormality observed during loading or transportation shall be immediately communicated to BRL.
- g. Transit insurance shall be arranged by the vendor wherever applicable.
- h. Rollers shall be unloaded at BRL using crane or forklift with nylon slings and wooden saddles only.
Note: Manual handling or dragging of rollers is strictly prohibited.
- i. Any transit damage observed at BRL shall be recorded during plant incoming inspection and NCR shall be raised.
- j. After receiving the roller refer to the 7.1.8 Plant Incoming Inspection of Roller.

7.1.8 Incoming Inspection of Roller at plant:

- a. On arrival of the roller at the plant, the Production team shall receive the material and verify the following documents:
- Approved PDI checklist (ROM/L4/005 & ROM/L4/006)
 - Delivery challan
 - NCR with corrective action (if applicable)
- b. The Production team (roller incharge) shall ensure proper unloading using approved lifting tools such as crane/forklift with nylon slings and wooden saddle supports.
Note: Direct contact of roller surface with floor or metal shall not be allowed.
- c. The roller shall be fixed on lathe machine and cleaned with sand blasting (1 cut or more if required) and visually jointly inspected by Engineering team and Production team (Roller incharge).
- d. Incoming inspection shall be carried out covering:
- Visual condition of chrome and surface
 - Damage during transportation
 - Shaft, journal, and bearing seat condition
 - Verification of roller ID / QR code

	Prepared By	Reviewed By	Approved By	Page number
Signature				Page 8 of 12
Name	Mr. Ripal Rana	Mr. Prabhakar P	Mr. Ajay Singh	
Designation	Sr. Executive	HOD - Hotend	Plant head	

- e. All inspection observations and measured values shall be recorded in the RMS web app (under remarks), including deviations (if any).
- f. In case of satisfactory condition:
 - Roller shall be accepted
 - Status updated in RMS as “READY FOR USE”
 - Roller should be transferred to designated storage locations.
- g. In case of deviation observed during incoming inspection:
 - If deviation is minor and acceptable, roller may be accepted with deviation notes and remarks recorded in RMS with approval of Plant head (refer flow chart).
 - If deviation is major and may affect roller performance or glass quality, roller shall be marked as **Rejected** and informed to vendor via mail (NCR raised and shared with the vendor).
- h. Any transportation-related damage shall be recorded separately, and responsibility shall be identified (vendor / transporter).
- i. Accepted rollers shall be stored in designated racks with proper identification and status tagging as per Section 7.1.1 – Roller Identification & Storage.

7.1.9 Roller Issue & Installation:

- a. Requirements for roller installation shall be raised by the Engineering team and Production team based on machine condition or planned replacement schedule.
- b. Before issue, the Rolling machine team shall verify in RMS that the roller status is “Ready for Use” and ensure all required approvals are completed.
- c. The issued roller shall be physically verified for:
 - Roller Number
 - Surface condition
 - Protection removal (only at installation area)
- d. Roller movement from storage to machine area shall be carried out using approved lifting tools such as crane or forklift with nylon slings and wooden saddles.

	Prepared By	Reviewed By	Approved By	Page number
Signature				Page 9 of 12
Name	Mr. Ripal Rana	Mr. Prabhakar P	Mr. Ajay Singh	
Designation	Sr. Executive	HOD - Hotend	Plant head	

- e. Installation shall be carried out by the Engineering team & Production team as per approved work instructions and safety procedures.
- f. Quality and Production teams should check the correlation between Roller and glass Ra value (for analysis purpose).
- g. Upon satisfactory results:
 - Roller installation shall be approved
 - RMS status shall be updated as “Running”
 - Installation record shall be completed and filed

7.1.10 Roller Scrap Control:

- a. A roller shall be considered for scrap when:
 - A roller is damage is beyond repair or completely worn out beyond certain diameter, it will be scrapped. And it is applicable for roller at all stages, if:
 - Roller is at BRL – Scrap it following Scrap approval process.
 - Roller is at Vendor – Take back to BRL and then scrap it following Scrap approval process.
- b. Scrap recommendation shall be initiated by the Roller incharge & Production HOD & move to red tag area.
- c. All scrap rollers shall be recorded in RMS web app, including.
 - Initiate corrective and preventive action.

7.2 Special Process Requirements
7.2.1 Process Requirements:

- a. Roller re-chroming, engraving, and shaft repair shall be treated as special processes.
- b. For Matt-Matt – Top and bottom Ra value is important.
- c. For Green House – Haze value is important.

7.3 Prevention & Detection Controls

7.3.1 PDI and Post-Engraving Inspection shall be conducted to capture defects or issues before it goes to production.

7.3.2 Use calibrated equipment and tools to prevent any measurement error.

	Prepared By	Reviewed By	Approved By	Page number
Signature				Page 10 of 12
Name	Mr. Ripal Rana	Mr. Prabhakar P	Mr. Ajay Singh	
Designation	Sr. Executive	HOD - Hotend	Plant head	

7.4 Controls Related to EHS

7.4.1 PPE MATRIX

			
Goggles	Safety Shoes	Hand Gloves	

8.0 EHS Compliance Obligations

8.1 Packing waste to be properly disposed of following the waste management practices.

8.2 Roller scrap should be done using the scrap disposal process.

9.0 Possible deviation & Impacts

9.1 Roughness or other parameter is not found as defined in specification during PDI - the glass surface may affect.

9.2 Roller sent by vendor without PDI confirmation – may introduce high risk of deviation.

9.3 After Confirmation also no PDI has been conducted by BRL team – delay in roller receiving

10.0 Proposed actions

10.1 Reject the Roller at stage of PDI and fill the NCR and ask vendor to correct the same.

10.2 No delivery will be taken and strict warning to be given to the vendor.

10.3 Written notification shall be given by the BRL to vendor for bypassing PDI process.

11. Reference documented information

11.1 Roller Inspection Plan - ROM/L4/009

11.2 Job card – ROM/L4/004

11.3 Roller PDI Checklist – ROM/L4/005

11.4 Post-Laser Engraving Inspection Checklist – ROM/L4/006

11.5 Non-Conformity Report – ROM/L4/007

11.6 Deviation report – ROM/L4/008

11.7 Annexure 1 – Roller management flow chart

11.8 Roller management portal – [Link](#)

	Prepared By	Reviewed By	Approved By	Page number
Signature				Page 11 of 12
Name	Mr. Ripal Rana	Mr. Prabhakar P	Mr. Ajay Singh	
Designation	Sr. Executive	HOD - Hotend	Plant head	

12. Revision History

Revision No.	Date	Description of changes
00	21.01.2026	Initial release
01	04.08.2026	Reference documented information added, flow chart annexure linked, PDI split into two activities (Post-Laser Engraving and PDI).

	Prepared By	Reviewed By	Approved By	Page number
Signature				Page 12 of 12
Name	Mr. Ripal Rana	Mr. Prabhakar P	Mr. Ajay Singh	
Designation	Sr. Executive	HOD - Hotend	Plant head	